**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# LESSON 056 · Selection: ELSE and Nested Conditions

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

## Do Now · Recall previous learning

Q1. What does selection do?

Q2. What does a structure diagram show?
A. The order instructions run
B. Every variable's value
C. The network's IP addresses
D. Parts and their hierarchy

Q3. “Doubling clock speed guarantees every program finishes in exactly half the time.” Correct this. Explain why.

Q4. A user runs several apps. Some work can be shared across cores. “Each extra core doubles every program's speed.” Correct this. When do more cores help?

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# I Do · Read new knowledge

## Key vocabulary

ELSE — An ELSE branch runs when its IF condition is false.

ELSEIF — An ELSEIF tests another condition after an earlier branch did not match.

Nested selection — An IF statement can be placed inside another to check a second condition.

Mutually exclusive branches — Only one matching route of an IF/ELSEIF/ELSE chain runs.

Algorithm — An ordered set of instructions to solve a problem.

Trace — Following instructions and recording changes to values or output.

## New learning

What is ELSE?: ELSE — An ELSE branch runs when its IF condition is false.

In this example: A school attendance award uses elseif; An ELSEIF tests another condition after an earlier branch did not match.

Follow the effect: An IF statement can be placed inside another to check a second condition. Explain how this changes the eventual program result.

Check understanding: Mutually exclusive branches is used for a specific reason: Only one matching route of an IF/ELSEIF/ELSE chain runs.

## Worked example

A school attendance award is being written. Explain how else and elseif are used to make the program work. [4 marks]

1. We are designing a school attendance award. Identify which data and decisions are essential.

2. ELSE [1] — An ELSE branch runs when its IF condition is false. [1].

3. ELSEIF [1] — An ELSEIF tests another condition after an earlier branch did not match. [1].

4. Two linked points for each technique give four mark points.

# We Do · Practise with guidance

### Guided 1

In a online shop discount levels, explain how else helps the program. [2 marks]

Worked method: ELSE [1] — An ELSE branch runs when its IF condition is false. [1].

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

### Guided 2

For a theme park eligibility, explain why elseif matters. [2 marks]

Worked method: ELSEIF [1] — An ELSEIF tests another condition after an earlier branch did not match. [1].

### Guided 3

When designing a competition medal bands, explain how nested selection and mutually exclusive branches should be used. [4 marks]

Worked method: Nested selection [1] — An IF statement can be placed inside another to check a second condition. [1]. Mutually exclusive branches [1] — Only one matching route of an IF/ELSEIF/ELSE chain runs. [1].

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

## Code First · Predict → Write → Test → Improve

Complete the comparison so the program reports insufficient storage.

```
freeGB = 4
neededGB = 6
if ... then
    print("Enough space")
else
    print("Not enough space")
endif
```

write

## Question 1

State two programming terms or techniques relevant to this lesson. [2 marks] Source: Original OCR J277-style teaching question; not an official past-paper question.

## Question 2

A cafeteria age prices needs to work correctly. Explain how elseif and nested selection would help. [4 marks] Source: Original OCR J277-style teaching question; not an official past-paper question.

## Question 3

A pupil is programming a parking charge levels. Explain the roles of else, nested selection and mutually exclusive branches in their solution. [6 marks] Source: Original OCR J277-style teaching question; not an official past-paper question.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

## Exit Ticket · Answer before checking

An ELSE branch runs when its IF condition is false.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

## Page 1 onwards · Do Now

Q1. It chooses a branch using a condition. Selection can be nested.

Q2. D — Parts and their hierarchy

Q3. More clock cycles per second. This may let the CPU process more instructions if other factors are similar.

Q4. More cores help when work can be split between them. Other CPU features and the task also affect the speed gained.

## I Do / We Do pages · Guided practice

Q1. ELSE [1] — An ELSE branch runs when its IF condition is false. [1].

Q2. ELSEIF [1] — An ELSEIF tests another condition after an earlier branch did not match. [1].

Q3. Nested selection [1] — An IF statement can be placed inside another to check a second condition. [1]. Mutually exclusive branches [1] — Only one matching route of an IF/ELSEIF/ELSE chain runs. [1].

## You Do pages · Code First

```
freeGB = 4
neededGB = 6
if freeGB >= neededGB then
    print("Enough space")
else
    print("Not enough space")
endif
```

Expected result: Not enough space

## You Do pages · Independent questions

Q1. ELSE [1]; Nested selection [1].

Q2. ELSEIF [1] — An ELSEIF tests another condition after an earlier branch did not match. [1]. Nested selection [1] — An IF statement can be placed inside another to check a second condition. [1].

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Q3. ELSE [1] — An ELSE branch runs when its IF condition is false. [1]. Nested selection [1] — An IF statement can be placed inside another to check a second condition. [1]. Mutually exclusive branches [1] — Only one matching route of an IF/ELSEIF/ELSE chain runs. [1].

### Exit Ticket · Model answer

True

## Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

**1. Copy the whole original question:**

**2. Write a complete new answer in GREEN PEN. Show working if needed:**